MODEL C / END-TO-END REPORT

Freight-cost movements

Freight and Brent inputs, direct quantiles and model comparisons

2005-2026 | 260 monthly observations | 8 features

Direct forecasts of a broad freight producer-price index. Gradient Boosting remains the H3 default; Random Forest is a retrospective comparator, not a promoted validation winner.

ASSESSMENT / READ BEFORE USE

Research / demo. H6 is a fallback; interval reliability and H12 stress results are weak.

Inside: data sources and preprocessing; features and targets; chronological splits; trained and selected models; validation, disruption and post-disruption comparisons; prediction outputs; graphs; reproducibility and limitations.

This is a formatted PDF edition of the packaged README. The figures are embedded from saved model visualizations. No models, datasets or forecasts were changed. Source documentation: 13 October 2026. PDF generated: 15 October 2026.

PDF bookmarks provide section navigation. Relative file links refer to the ready-to-go package; keep this PDF at its root to retain those references.

Updated 13 October 2026 from the saved training_v1 data, code, selection records and evaluation outputs. Documentation only; no models were retrained or promoted.

Summary And Readiness

Model C predicts **future log changes in a deep-sea freight producer-price index**, with seven quantiles at H3/H6 and additional **H12 reporting-only**. It converts these changes into percentage movement and an index level for interpretation. This is supervised direct-horizon probabilistic regression, not a recursive/unsupervised forecast.

The saved default is **Gradient Boosting H3**, **no-change baseline H6**, and **Gradient Boosting H12**. Random Forest is the strongest retrospective H3 disruption comparator, but it failed the validation baseline comparison; it is not the pre-test-selected default.

Results are mixed: H3 median MAPE is about **3.70% during disruption**, but the selected model's interval coverage is only **87.50%**, below the 90% requirement. H6 remains a fallback and H12 disruption coverage is **25%**. This is a research/demo with visible limitations, not production acceptance.

Data, Sources And Meaning

Item	Prepared dataset used here
Input	model_c_freight_training_2005_2026.csv
Frequency / coverage	Monthly, January 2005-August 2026
Size	260 reference months; 67 columns, of which eight are estimator features
Freight source	US Bureau of Labor Statistics deep-sea freight producer-price index, obtained from FRED PCU483111483111
Fuel signal	Brent crude-oil monthly USD/barrel price from World Bank Commodity Price Data / Pink Sheet
Metadata	Preparation manifest, column dictionary

The freight index measures price movements relative to its reference base, not an absolute freight quote in GBP, USD or GBP/kg. Brent is a direct oil price used to construct a historical fuel-cost signal. **The model does not predict Brent prices**, and it is not trained on titanium import values. Its relationship to titanium is as a broad freight-cost proxy, not a titanium-specific route tariff.

The source files have longer history; 2004 is used for feature warmup, but labelled/exported rows start in 2005. The prepared snapshot ends in August 2026 where both inputs are aligned. The raw source CSV/workbook remain in the sibling Model C Data directory for provenance; inference/training read local data/ and do not require those raw files.

Preparation And Targets

Inputs are aligned monthly, historical changes and volatility are computed, calendar/timing evidence is retained, and future observed freight values are shifted to origin rows as supervised labels. Early warmup is separated from exported training history. Final not-yet-observed targets remain blank; they are not filled with predictions or fabricated actuals.

Quantity	Formula
Historical change over k months	ln(freight_index[t] / freight_index[t-k])
Fitted future target at h	ln(freight_index[t+h] / freight_index[t])
Predicted percentage movement	100 * (exp(predicted_log_change) - 1)
Predicted index level	freight_index[t] * exp(predicted_log_change)

t-12 means the **single observation twelve months earlier**, not the sum/mean of the preceding twelve values. H3 means the target month three months ahead, not an average of the next three months.

Horizon	Fitted target	Supporting observed outcome
H3	target_freight_log_change_h3	target_freight_index_h3, target_freight_pct_change_h3, target_month_h3
H6	target_freight_log_change_h6	target_freight_index_h6, target_freight_pct_change_h6, target_month_h6
H12	target_freight_log_change_h12	target_freight_index_h12, target_freight_pct_change_h12, target_month_h12

Labels are calculated from genuine later observations for historical supervised training. They are never predictor inputs. Timing/eligibility fields control available labels; assigned assumed-release flags are not random synthetic market data and are not estimator features.

Eight Features Used

Exact feature list: results/run_manifest.json.

Input feature	Calculation / role
freight_log_change_1m	Current versus previous-month log change; recent movement
freight_log_change_2m	Current versus two-month-earlier log change
freight_log_change_3m	Current versus three-month-earlier log change
freight_log_change_6m	Current versus six-month-earlier log change
freight_log_change_12m	Current versus twelve-month-earlier log change; annual movement
freight_volatility_6m	Population standard deviation of latest six monthly freight log returns
freight_volatility_12m	Population standard deviation of latest twelve monthly freight log returns
brent_log_change_1m	ln(Brent[t] / Brent[t-1]); historical oil/fuel-cost movement

Volatility includes the current observed month, uses ddof=0 and is not annualized. Only these eight inputs enter the fitted regressors. Freight/Brent availability dates, target dates, future targets, split labels and verification flags do not.

Feature importance is not causal evidence. Ridge coefficients, tree impurity importance and Gradient Boosting P50 importance are different diagnostic measures. The no-change baseline has no learned feature mapping and no learned feature importance.

Train, Validation And Test Split

This is an expanding chronological evaluation, not a random split. Settings were fixed before the disruption evaluation. Each historical fit uses only past outcomes assumed available at its issue date. Later fits may include earlier test outcomes after publication; this simulates ongoing refitting rather than one frozen 2019 estimator throughout testing.

Stage	Reference origins	Actual evaluated size / purpose
Initial history	January 2005-December 2014	120 reference months; eligible fit counts differ by horizon/timing
Main validation	2015-2019	56 H3, 53 H6 and 47 H12 eligible outcomes known before 1 January 2020
Eligible validation origins	H3 ends August 2019; H6 May 2019; H12 November 2018	Late origins cannot enter selection before their targets are known
Pretest bridge	Late 2019 origins, horizon-dependent	Excluded from selection; completed predictions can support later recovery diagnostics
Primary disruption test	January 2020-December 2021 reference months	24 forecasts per model/horizon; COVID-era freight disruption
Separate issue-date test	Assumed forecast issue dates in January 2020-December 2021	Also 24 per model/horizon, but boundary reference months differ; not interchangeable with the primary table
Post-disruption monitoring	January 2022 onward	H3 ends May 2026 (53); H6 February 2026 (50); H12 August 2025 (44)
Latest saved origin	August 2026	Assumed issue date 30 September 2026

The primary test labels **origins**, not outcomes. H3 disruption targets extend through March 2022, H6 through June 2022 and H12 through December 2022. Post targets end August 2026.

The user chose initial history 2005-2014 and validation 2015-2019. The broader specification's 2012-2019 validation was not fully implemented: 2012-2014 remain initial training, not a separate early backtest.

Calibration And Timing

The latest **24 eligible labelled origins** are held out for chronological quantile calibration. Earlier proper-training rows are additionally purged so their targets were available at the first calibration origin's assumed issue date. Both model and baseline use the same holdout/purge protocol but their own errors.

Held-out P05/P95 miss scores determine a nonnegative conformal expansion. Intermediate quantiles shift proportionally around P50; the median remains unchanged and outputs remain ordered. Overlapping outcomes and nonstationary time series do not satisfy a simple exchangeability guarantee, so nominal 90% coverage is tested, not promised.

Source observations are assumed usable at the end of the following month. Actual historical release dates and revision vintages are unverified; the preparation manifest records zero strictly point-in-time verified rows.

Models Compared And Saved Default

Algorithm	Learning and quantile distribution
Ridge + residual quantiles	Standardized linear regression plus empirical training-residual quantiles
Quantile Random Forest	Trees fitted to log changes; quantiles from historical target values in reached leaves, equal total weight per tree
Quantile Extra Trees	More randomized splits with the same leaf-distribution quantile method
Gradient Boosting quantiles	Separate quantile-loss regressor for each of seven probabilities
No-change baseline	Median future log change zero; historical no-change errors provide uncertainty

There are **15 saved bundles**: five approaches x three horizons. A Gradient Boosting bundle contains seven quantile estimators; it is not one multi-output estimator. H6/H12 predict their own labelled horizons, not recursive H3 outputs.

Horizon	Saved default	Selection reason / artifact
H3	Gradient Boosting	Passed validation coverage/MIS gates; model_c_h3.joblib
H6	No-change baseline	No ML candidate passed validation gates; model_c_h6.joblib
H12	Gradient Boosting	Family selected from H3/H6 comparison only; reporting-only; model_c_h12.joblib
H3 comparator	Random Forest	Stronger historical disruption result, not promoted after testing; model_c_h3.joblib

Selection record. H3/H6 selection prefers coverage >=90%, no missing forecasts and MIS better than baseline; otherwise it retains the baseline and reports the best ML candidate. Ridge has the lowest H6 ML validation MIS, but fails coverage. H12 scores are excluded from choosing the model family.

Evaluation Metrics And Recovery

Metric	Meaning
MAE / RMSE in index points	Absolute / squared-error-sensitive median index error
MAPE / sMAPE	Percentage deviation of median index forecast; not classification accuracy
MAE/RMSE log change	Error on the actual fitted target scale
Pinball loss	Mean quantile loss over all seven outputs
P05-P95 coverage	Fraction inside the nominal 90% interval
MIS / baseline	Width and miss penalties at alpha 0.10, in log-change units; below 1 is better
Recovery	Six completed, assumed-published outcomes achieve >=90% coverage after disruption, within the implemented six-month deadline

With only six completed observations, >=90% coverage requires **six of six** inside the interval. The recovery definition and dates are in recovery_summary.csv; it differs from A's origin-based recovery. Probability gates require coverage and MIS superiority with no missing outputs; full disruption gates additionally require recovery. H12 is reporting-only. Baseline cannot beat its own MIS, so its acceptance flag is not a separate self-comparison failure.

Validation Comparison: All Approaches

Algorithm	H3 coverage	H3 MIS / baseline	H3 MAPE	H6 coverage	H6 MIS / baseline	H6 MAPE
Ridge	92.86%	1.0647	2.87%	86.79%	0.9778	5.29%
Random Forest	94.64%	1.1264	2.36%	94.34%	1.1087	4.45%
Extra Trees	92.86%	1.0985	2.24%	98.11%	1.0076	4.42%
Gradient Boosting	94.64%	0.9887	2.32%	88.68%	1.0362	4.33%
No-change	98.21%	1.0000	2.39%	100.00%	1.0000	4.36%

Validation metrics. Gradient Boosting's H3 advantage in MIS is small, around 1.1%, not overwhelming proof of superiority.

Primary Disruption Comparison: All Approaches

Algorithm	H3 coverage	H3 MIS / baseline	H3 MAPE	H3 full gate	H6 coverage	H6 MIS / baseline	H6 MAPE
Ridge	91.67%	0.7997	3.93%	Fail recovery	75.00%	1.1103	8.56%
Random Forest	95.83%	0.6829	3.72%	Pass	62.50%	1.2802	7.34%
Extra Trees	91.67%	0.7247	3.90%	Fail recovery	62.50%	1.1751	7.49%
Gradient Boosting	87.50%	0.8450	3.70%	Fail coverage/recovery	70.83%	0.9370	7.31%
No-change	79.17%	1.0000	4.47%	Reference	66.67%	1.0000	8.38%

Random Forest improves H3 disruption MIS by about 31.7% versus baseline, but was worse in validation. Picking it after viewing the test would reuse the test for selection. No H6 ML model reaches 90% disruption coverage.

Selected Default: Validation, Disruption And Post-Disruption

Period	Default / H	N	MAE index	RMSE index	MAPE	Coverage	MIS	MIS / baseline
Validation	Gradient Boosting H3	56	6.2810	7.8792	2.32%	94.64%	0.1510	0.9887
Validation	No-change H6	53	11.7377	13.5448	4.36%	100.00%	0.2528	1.0000
Validation	Gradient Boosting H12	47	26.3448	30.5668	9.45%	80.85%	0.4230	1.0835
Disruption	Gradient Boosting H3	24	11.9265	14.7679	3.70%	87.50%	0.2438	0.8450
Disruption	No-change H6	24	29.5177	36.2870	8.38%	66.67%	0.4880	1.0000
Disruption	Gradient Boosting H12	24	68.7632	84.5234	16.68%	25.00%	1.5668	1.1148
Post	Gradient Boosting H3	53	28.2190	33.2080	6.52%	94.34%	0.3687	1.0147
Post	No-change H6	50	33.6923	43.9685	7.62%	84.00%	0.4799	1.0000
Post	Gradient Boosting H12	44	31.7735	40.2212	7.33%	90.91%	0.6262	1.1139

Source: evaluation_metrics.csv. H12 is not an acceptance decision. All displayed groups have complete predictions and observed targets. Separate issue-date test results are labelled separately in that file; for example selected H3 coverage is 83.33%, not the primary test's 87.50%.

Interpretation: selected H3 central forecasts are fairly close during disruption, but under-cover uncertainty. Post H3 coverage recovers while MIS is slightly worse than baseline, and MAPE rises to 6.52%. H6's fallback intervals under-cover both later periods. H12 is particularly weak during disruption. Plausible positive index levels and small median errors do not establish calibrated future risk.

Ground Truth, Errors And Loss Curves

Compare **actual_freight_index** with **p50_freight_index** (also predicted_freight_index) in the saved prediction records. Compare actual_log_change with p50_log_change for fitted-target error. Origin_freight_index is the known starting point, not future actual.

Signed error = predicted - actual; absolute_pct_error = 100 * abs(predicted - actual) / actual. An average MAPE of 3.70% does not imply every month is within 3.70%, nor a guaranteed 96.30% probability of being correct.

Validation records | Disruption records | Post records | Separate issue-date test.

Training/validation diagnostics use **raw uncalibrated pinball loss**, not classification accuracy. Ridge varies chronological training size; forests vary tree count; Gradient Boosting shows staged estimates. These are not all epoch-based curves. Initial proper-training predictions are in-sample diagnostics and do not determine readiness; the diagnostic validation subset is 2015-2017. Curve values, in-sample metrics.

Latest Selected Forecasts

Reference observation: **August 2026**, index **518.6170**; assumed issue date **30 September 2026**. Horizons count from the observation, not the current computer month.

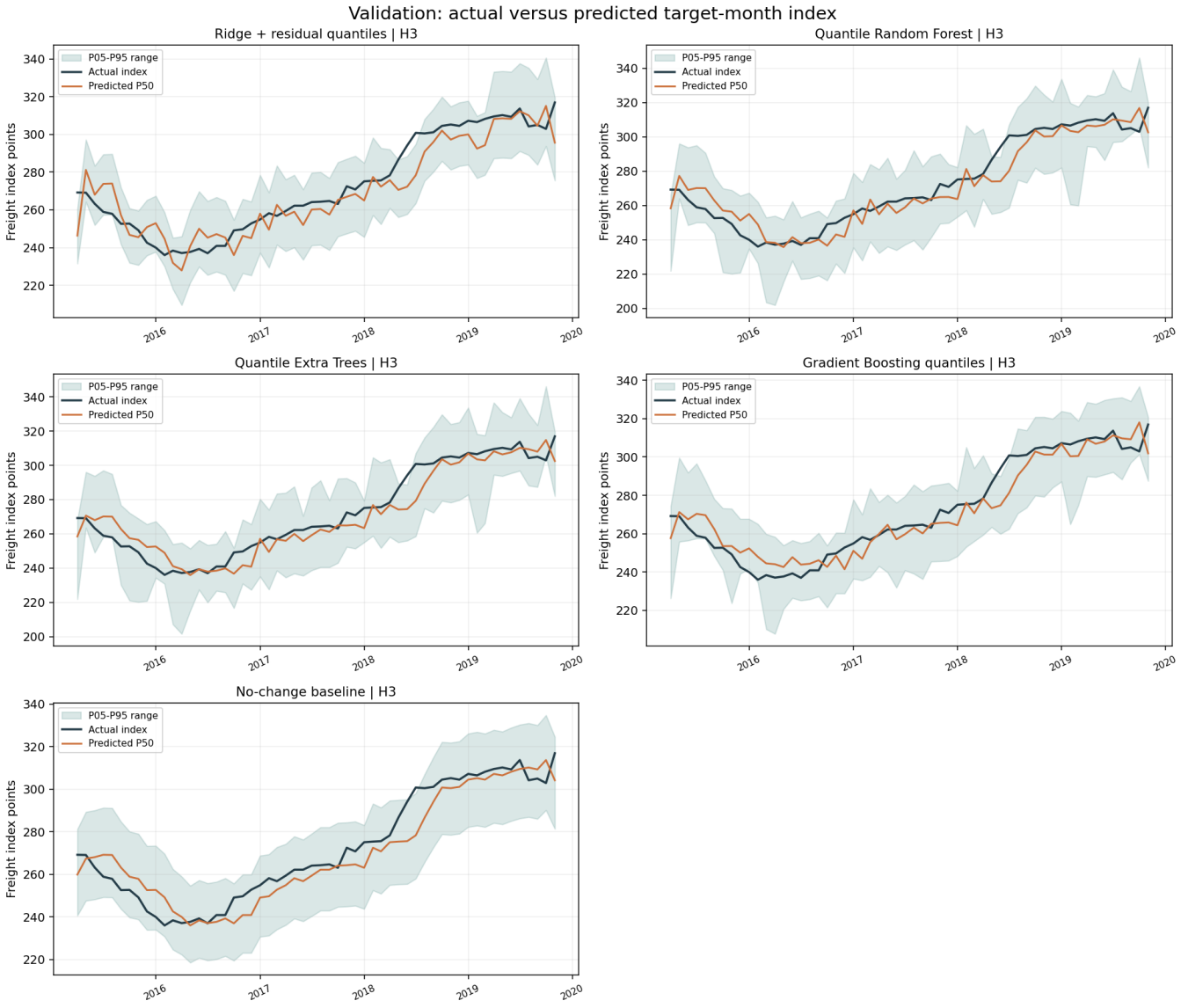
Horizon / target	Selected model	P05 index	P50 index	P95 index	P50 movement
H3 / November 2026	Gradient Boosting	478.4597	530.9941	582.8774	+2.39%
H6 / February 2027	No-change	457.2769	518.6170	611.7665	0.00%
H12 / August 2027	Gradient Boosting	374.6107	542.5775	725.4727	+4.62%

Selected forecast CSV, all models CSV, response JSON. Outputs include all seven ordered quantiles in log-change, percentage and index units, model, horizon, target/reference months and reporting status.

The H6 constant median is intentional no-change behavior, not a display bug; its interval is still nonconstant uncertainty. H12's large interval and weak stress evidence should be visible. These forecasts have no future ground truth in the snapshot. They do not generate monthly forecasts for every intervening month or an exact shipping bill.

Figure 1. Validation actual versus predicted, H3

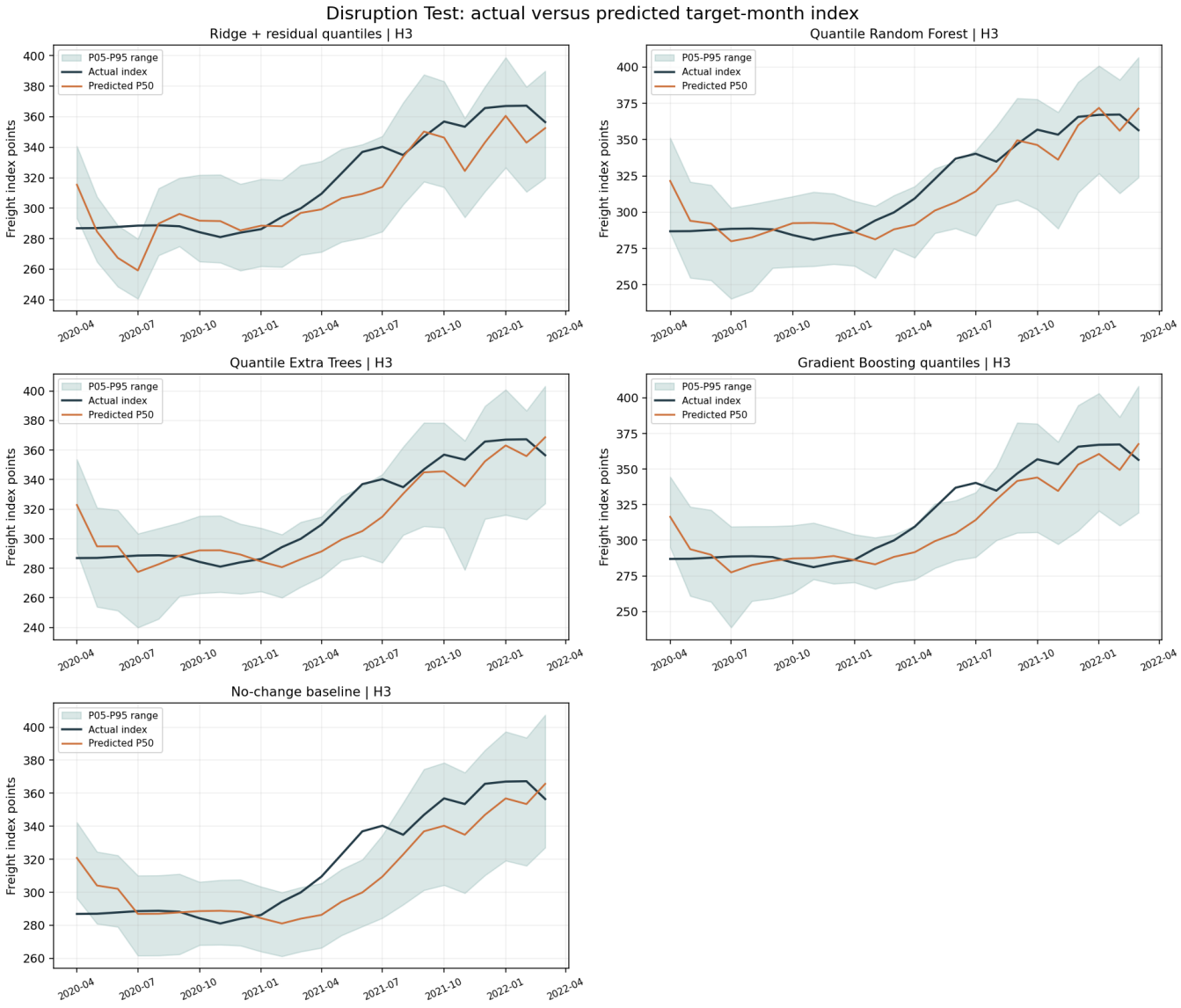
Saved visualization / Graphs And Dashboard



Source: visualizations/actual_vs_predicted_validation_h3.png

Figure 2. Disruption actual versus predicted, H3

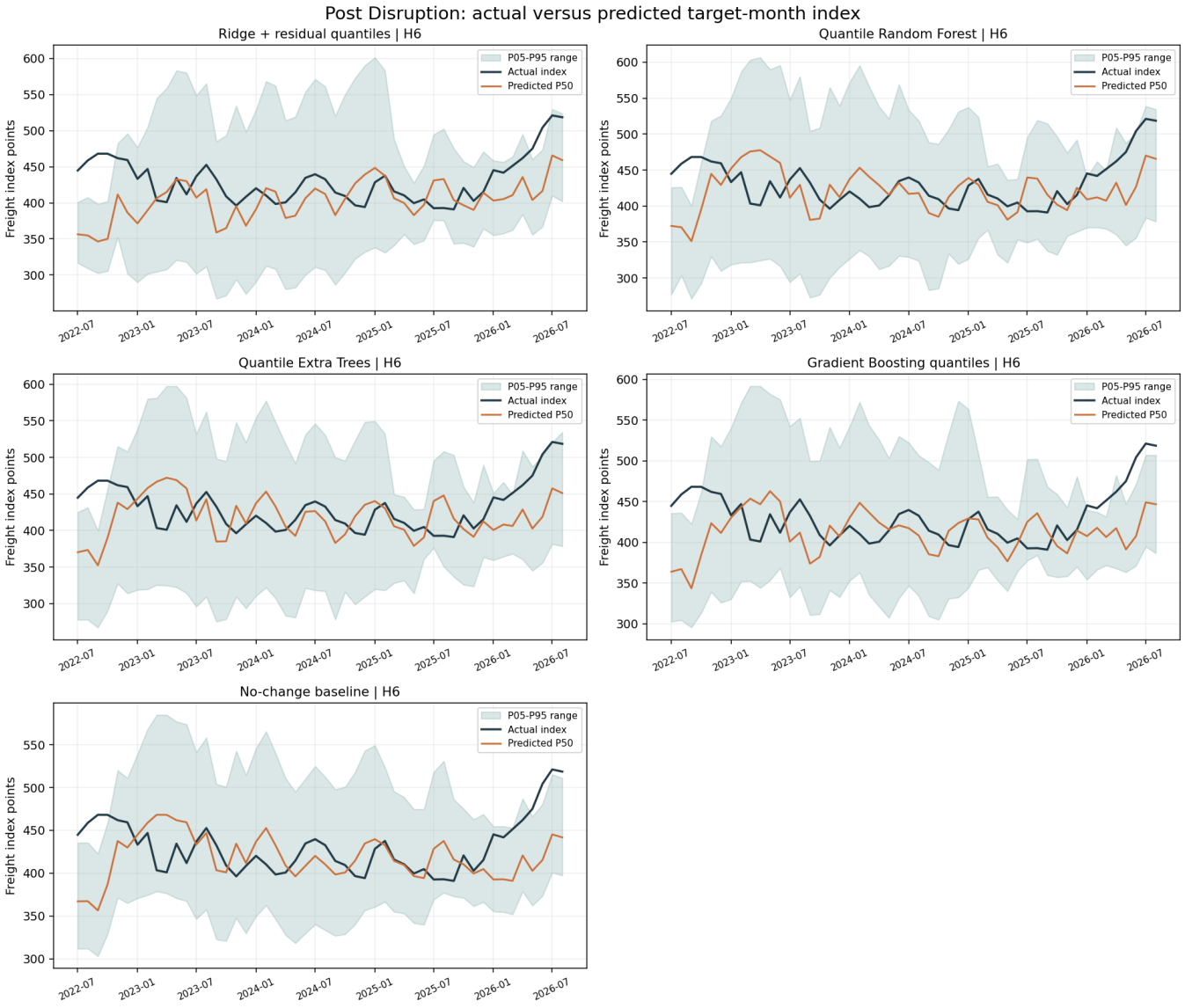
Saved visualization / Graphs And Dashboard



Source: visualizations/actual_vs_predicted_disruption_test_h3.png

Figure 3. Post-disruption actual versus predicted, H6

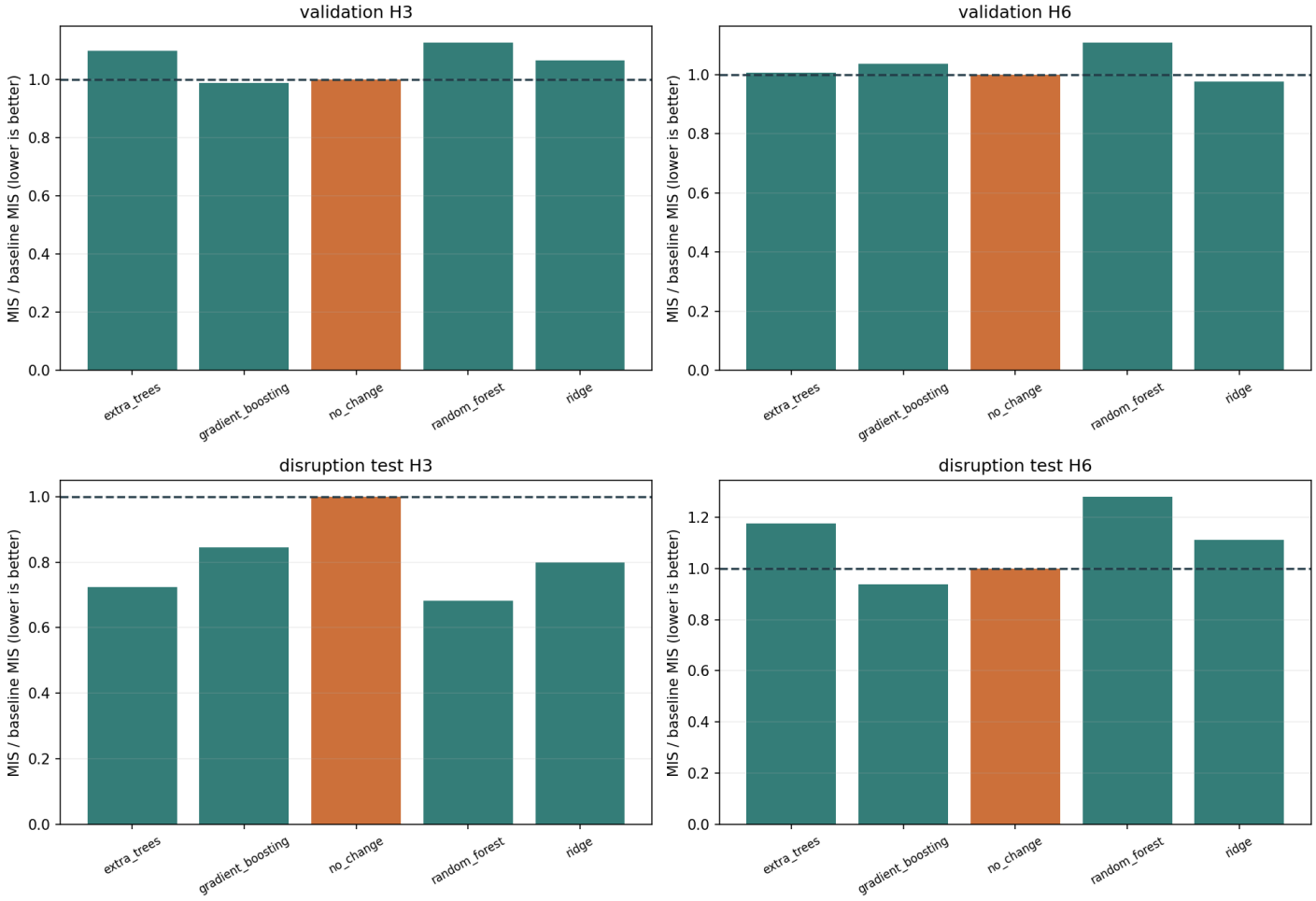
Saved visualization / Graphs And Dashboard



Source: visualizations/actual_vs_predicted_post_disruption_h6.png

Figure 4. MIS comparison

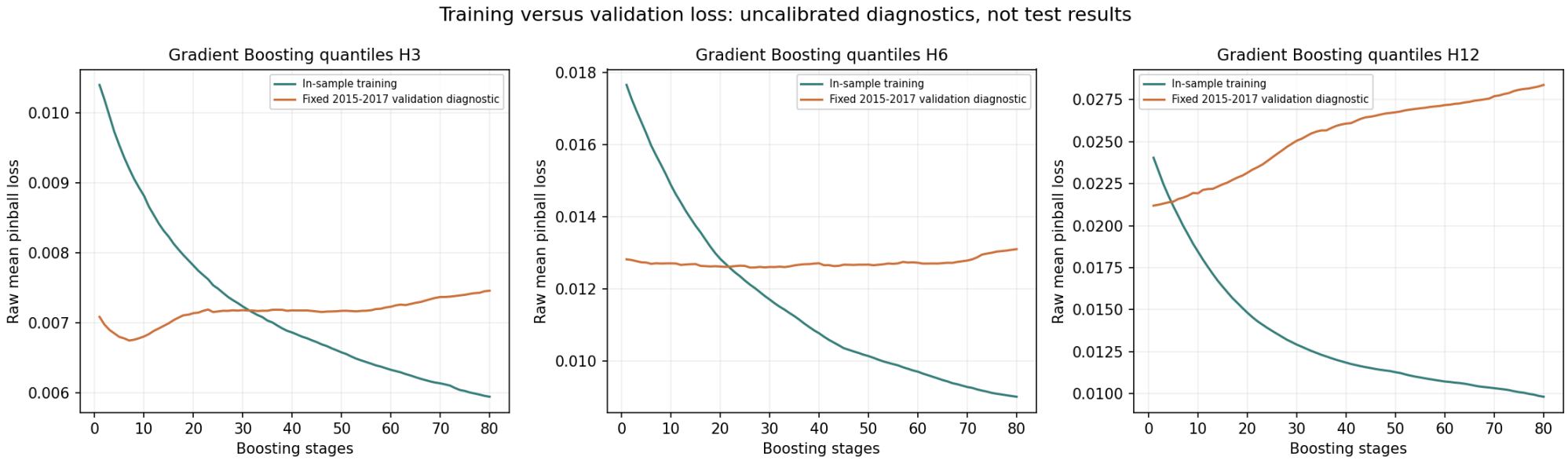
Saved visualization / Graphs And Dashboard



Source: visualizations/model_mis_comparison.png

Figure 5. Gradient Boosting training and validation diagnostics

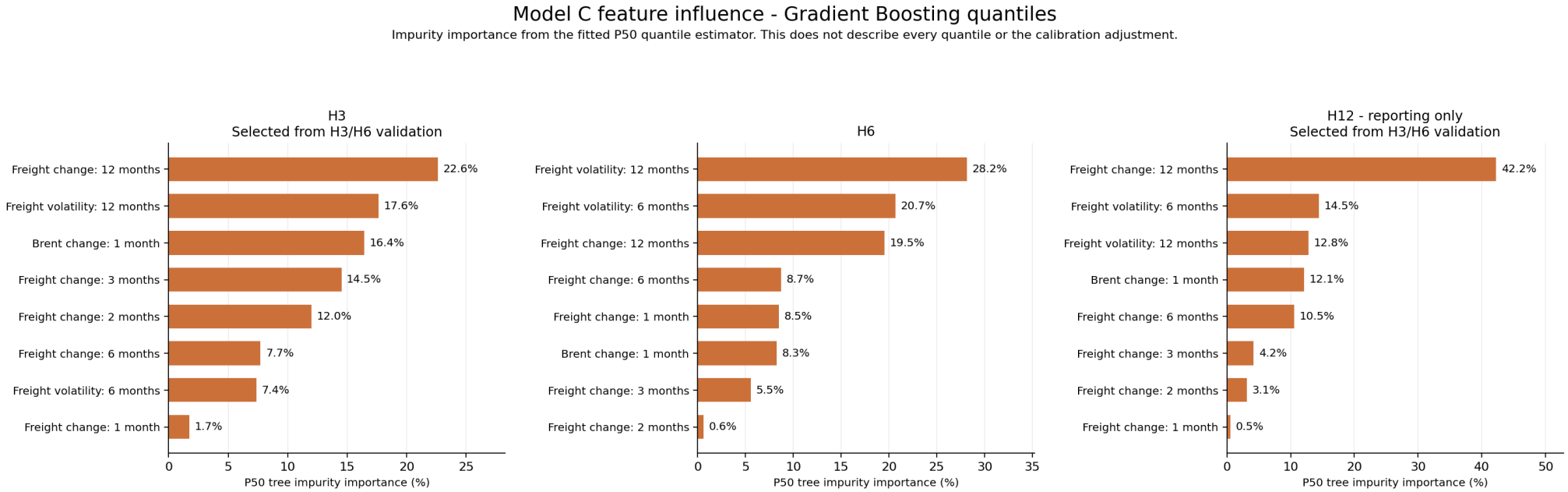
Saved visualization / Graphs And Dashboard



Source: visualizations/train_validation_loss_gradient_boosting.png

Figure 6. Gradient Boosting feature influence

Saved visualization / Graphs And Dashboard



Source: visualizations/feature_importance_gradient_boosting.png

Interactive dashboard | Recovery chart | Random Forest feature influence | Ridge diagnostics.

Other H3/H6/H12 actual-versus-predicted panels and model-specific loss/importance charts are in visualizations/. GitHub renders PNGs; download HTML for local interaction.

Folder Map And Commands

Location	Contents
data/	Package-local prepared CSV, preparation manifest and column dictionary
code/	Training/inference, independent verification, plotting and local-data verification
models/algorithm/	Fifteen fitted bundles and calibration metadata
predictions/	Out-of-sample actual/predicted records, in-sample diagnostics and latest forecasts
results/	Metrics, recovery, model selection, loss curves, feature influence and manifests
visualizations/	PNG charts and standalone HTML dashboard

Run from training_v1:

```
python -m pip install -r requirements.txt
python code/verify_local_data.py
python code/train_model_c.py --predict-only
```

Local verification reproduces **28 latest quantile-index values**, including the H3 Random Forest comparator, without fitting. Predict-only rewrites latest forecast output files but not historical evaluation. The combined root's VERIFY_PACKAGE.cmd verifies all models and the UI. The package reads local data/; no sibling Train directory is required.

Running the training script without flags performs fitting; it refuses to overwrite an already completed run. Preserve the selected evidence and use a new run folder for experiments. Rendering/plotting flags may write report artifacts. Load only trusted joblib files.

Remaining Limitations And Next Work

Obtain verified historical release/vintage information, complete the specification's FBX spot-rate comparison for 2020-2021, address the missing early-validation study, and test interval stability in a genuinely prospective holdout. Consider new genuine freight-route/fuel signals or models, selected on validation rather than the disruption leaderboard.

The BLS index is a broad US contract-oriented proxy, not a UK titanium route quote. H6 has no accepted ML candidate; H3/H12 lack full acceptance. Conformal nominal coverage is not guaranteed under changing time-series regimes. Saved reproduction proves code consistency, not forecast reliability. No comparison of A MIS, B Brier and C MAPE establishes one overall best model.

See the root source/reuse review before publication. Pinned versions match the verified local environment; installation in a fresh environment remains untested. Research/demo status and all limitations are unchanged.